C AROUND US

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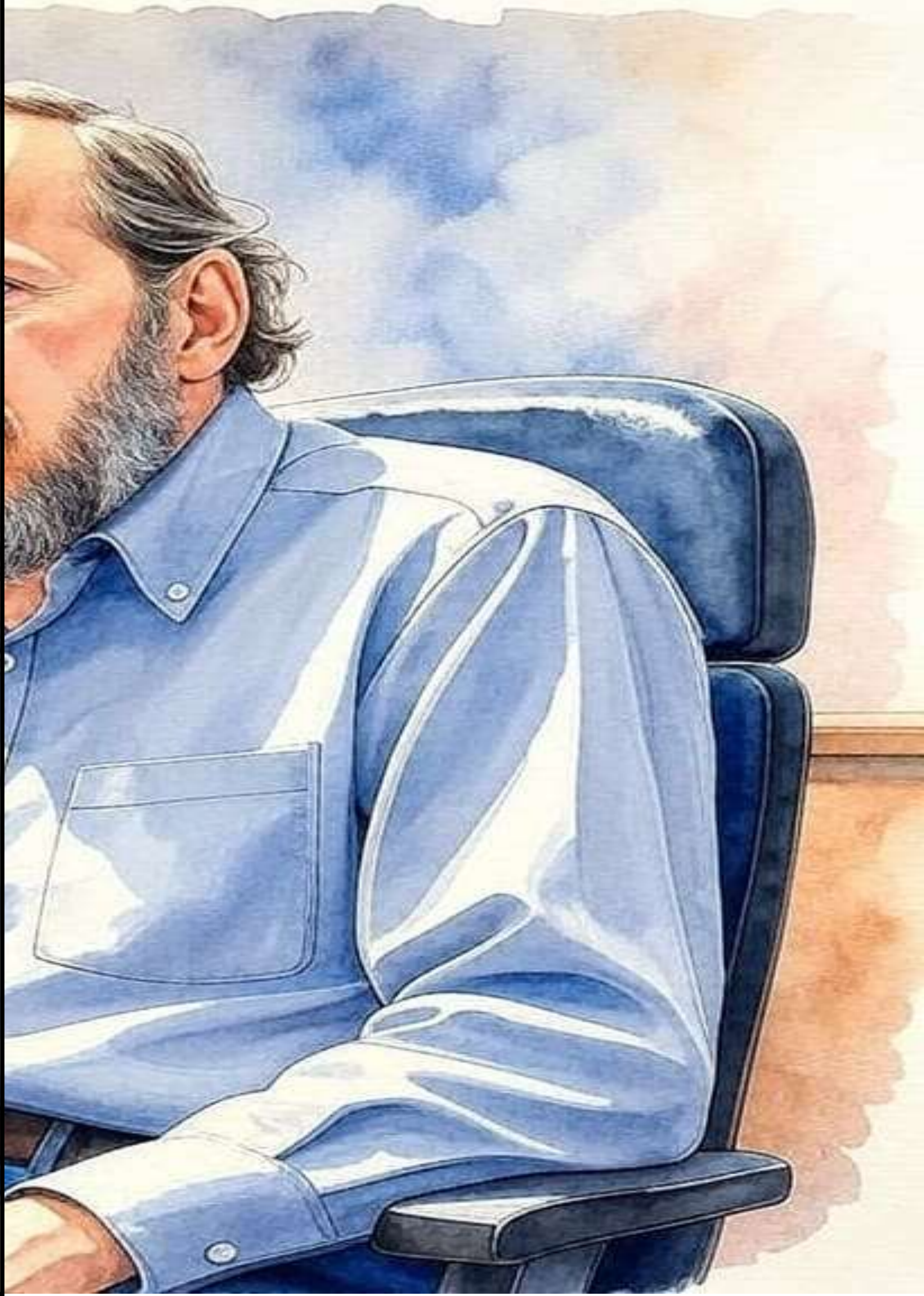
INTRODUCTION TO C

- C was developed by Dennis Ritchie in 1972.
- It was created at Bell Labs.
- C is a general-purpose, structured programming language.
- It provides low-level memory access and direct hardware control.
- C is fast, efficient, and widely used in system programming.
- Many modern languages like C++, Java, and C# are influenced by C.

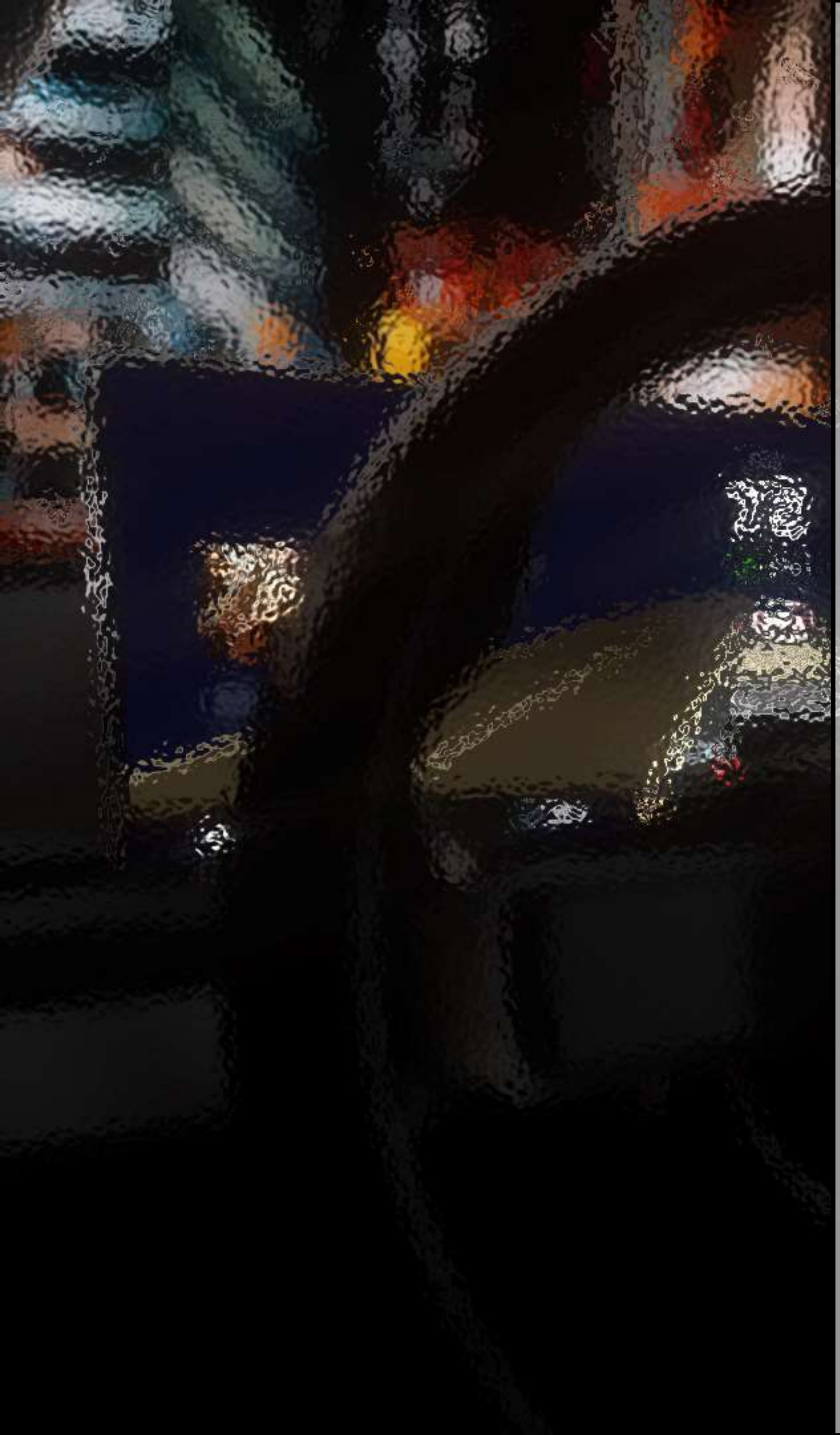


C IN EMBEDDED SYSTEMS & EVERYDAY DEVICES

- **ATMs :** C is used to securely verify PINs, check account balances, and process transactions in real time. It ensures fast and reliable banking operations.
- **Traffic Lights :** Traffic control systems use C to manage signal timing, read vehicle sensors, and ensure road safety through real-time monitoring.
- **Washing Machines :** C controls wash cycles, water levels, motor speed, and temperature through embedded microcontrollers.
- **Lift Systems :** Lift controllers programmed in C read floor buttons and safety sensors to ensure smooth and secure movement.
- **Alarm Clocks & Digital Devices :** C helps maintain accurate time tracking and triggers alarms precisely when required.

C IN OPERATING SYSTEMS

- Major operating systems like Linux are largely written in C.
- Parts of Windows use C for core system functions.
- C helps manage memory allocation, processes, and file systems.
- It enables direct communication between software and hardware.
- C provides speed and efficiency required for system-level programming.



C IN AUTOMOBILE SYSTEMS

- Engine Control Units (ECU) use C for real-time vehicle control.
- Controls fuel injection, ignition timing, and engine performance.
- Used in safety systems like Anti-lock Braking System (ABS).
- Airbag control systems rely on C for quick response.
- Ensures high reliability, speed, and safety in vehicles.

C AROUND US

C is a powerful, general-purpose programming language widely used in embedded systems, operating systems, and hardware-level programming. It is known for speed, efficiency, and direct hardware control.

MONEY AND BANKING

- ATMs use C for secure PIN and balance verification.
- Card machines process transactions in real time.
- Ensures fast, safe, and 24×7 operation.

TRAFFIC CONTROL

- Controls red, yellow, and green signals.
- Reads vehicle and pedestrian sensors.
- Prevents accidents through safety checks.

HOME APPLICATIONS

- Washing machines control wash and spin cycles.
- TVs use C for firmware and smooth performance.
- Lift systems manage floor selection and door safety.
- Alarm clocks track time and trigger alerts accurately.

ADVANCED APPLICATIONS

OPERATING SYSTEMS

- Major operating systems like Linux are largely written in C.
- Parts of Windows use C for system-level functions.
- C helps manage memory, processes, and hardware communication.
- It provides speed and direct control over system resources.

AUTOMOBILE SYSTEMS

- Engine Control Units (ECU) use C for real-time control.
- Controls fuel injection, braking, and engine timing.
- Ensures safety, speed, and reliability.
- Used in ABS and airbag systems.

WHY C ?

- ⚡ Fast and efficient
- 🔧 Close to hardware
- 🌍 Portable
- 🛡 Reliable and time-tested
- 📚 Foundation for other languages

C remains the backbone of modern computing and embedded technology.

DIGITAL BANKING SYSTEM

- Banking system is a **real-world application** used daily
- Demonstrates **practical use of C programming**
- Uses **structures** to store account details
- Implements **file handling** for data storage
- Includes **security using PIN verification**
- Demonstrates **transactions like deposit, withdraw, transfer**
- Shows **menu-driven program implementation**
- Helps understand **logic building and validation**

BANKING SYSTEM FEATURES

```
=============================================
         DIGITAL BANK CONSOLE PORTAL
=============================================
1. Open Account
2. Login
3. Exit
=============================================
Choose an option: 1


Open New Account
----------------
Full Name: Riddhi Parate
Email: riddhiparate04@gmail.com
Phone (10 digits): 1234567890
Create Password: ********
Confirm Password: ********
Create 4-digit PIN: ****
Confirm PIN: ****
Initial Deposit (minimum 0): 200000

Account created successfully.
Account Number: 1000011392
You can now login from the main menu.

Press Enter to continue...


=============================================
         DIGITAL BANK CONSOLE PORTAL
=============================================
1. Open Account
2. Login
3. Exit
=============================================
Choose an option:
```

C Version

Create Account
Login System
Deposit / Withdraw
Transfer Money
PIN Verification
Transaction History
File Storage

HTML Version

Same logic implemented in UI
Better user interface
Realistic banking experience
Local storage used

ALARM CLOCK FEATURES

- Set alarm time (hours and minutes)
- Displays current system time
- Compares current time with alarm time
- Triggers alarm when time matches
- Uses loops for continuous time checking
- Uses conditional statements for comparison
- Simple and user-friendly interface
- Demonstrates real-time application of C programming
- Extended version created using HTML, CSS and JavaScript
- Shows practical use of time-based logic

WHY C ?

Fast and Efficient: Produces high-speed programs with low memory usage.

Close to Hardware: Allows direct access to memory and hardware components.

Portable: Programs can run on different systems with minimal changes.

Foundation Language: Influenced modern languages like C++, Java, and C#.

Backbone of Systems: Core parts of Linux and other system software are written in C.

THANK YOU